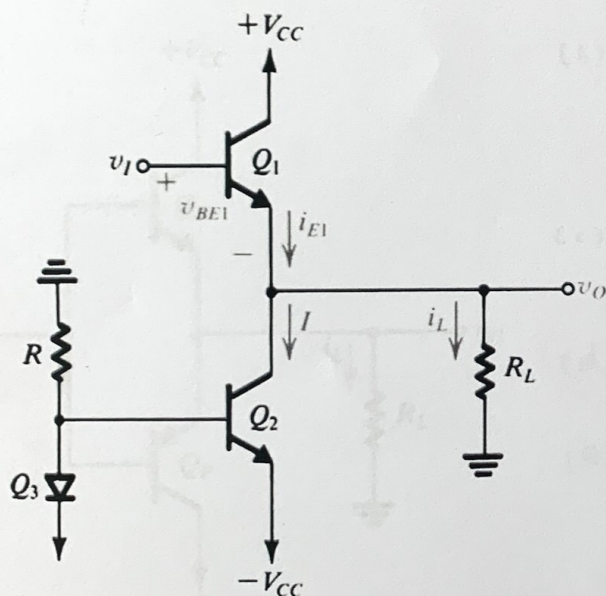


作业

12.3 For the emitter follower of Fig. 12.3, let $V_{CC} = 10\text{ V}$, $I = 100\text{ mA}$, and $R_L = 100\ \Omega$. If the output voltage is an 8-V-peak sinusoid, find: (a) the power delivered to the load; (b) the average power drawn from the supplies; (c) the power-conversion efficiency. Ignore the loss in Q_3 and R .

Ans. 0.32 W; 2 W; 16%



$$(a) P_L = \left(\frac{8}{\sqrt{2}}\right)^2 \cdot \frac{1}{100} = 0.32\text{ W}$$

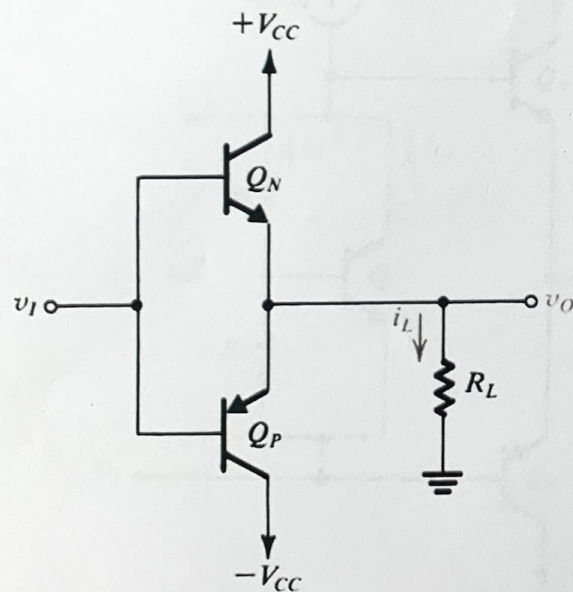
$$(b) i_L: -80\text{ mA} \sim 80\text{ mA}, \therefore i_{E1}: 20\text{ mA} \sim 180\text{ mA}, \text{平均值为 } 100\text{ mA}$$

$$P_S = 2V_{CC} \cdot 100\text{ mA} = 2\text{ W}$$

$$(c) \eta = \frac{0.32\text{ W}}{2\text{ W}} = 16\%$$

12.4 For the class B output stage of Fig. 12.6, let $V_{CC} = 6\text{ V}$ and $R_L = 4\ \Omega$. If the output is a sinusoid with 4.5-V peak amplitude, find (a) the output power; (b) the average power drawn from each supply; (c) the power efficiency obtained at this output voltage; (d) the peak currents supplied by v_i , assuming that $\beta_N = \beta_P = 50$; and (e) the maximum power that each transistor must be capable of dissipating safely.

Ans. (a) 2.53 W; (b) 2.15 W; (c) 59%; (d) 22.1 mA; (e) 0.91 W



$$(a) \hat{V}_o = 4.5\text{ V}, \quad \hat{i}_o = \frac{4.5}{4} = 1.125\text{ A}$$

$$P_L = \left(\frac{4.5}{\sqrt{2}}\right)^2 \cdot \frac{1}{4} = 2.531\text{ W}$$

(b) i_N 为半个正弦波

$$\therefore P_{S,N} = \frac{1}{\pi} \cdot \hat{I} \cdot V_{CC} = \frac{1}{\pi} \cdot 1.125 \cdot 6 = 2.15\text{ W}$$

$$(c) \eta = \frac{2.531}{2.15 \times 2} = 58.9\%$$

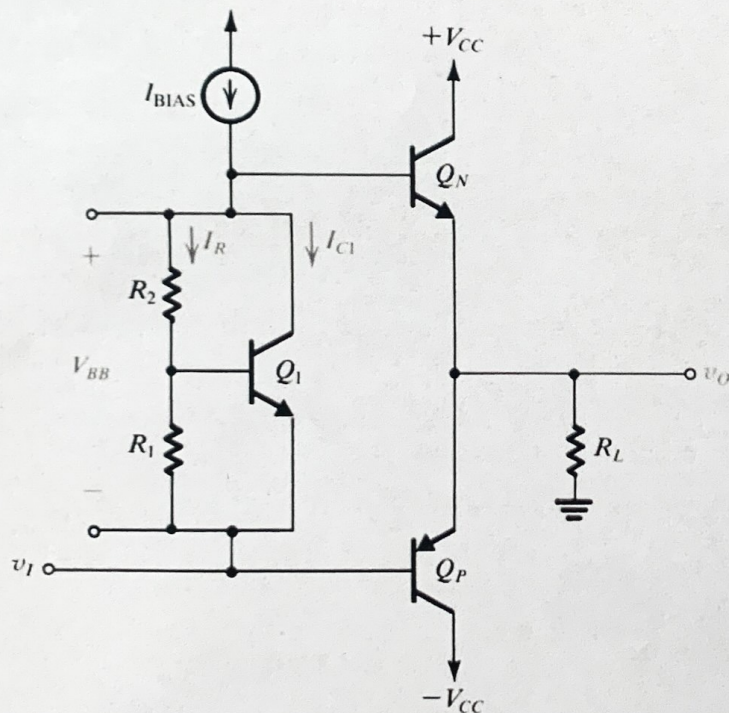
$$(d) \frac{1.125}{\sqrt{2}} = 22.1\text{ mA}$$

(e) 最大功耗值位于 $\hat{V}_o = \frac{2V_{CC}}{\pi} = 3.82\text{ V}$ 时, $3.82 < 4.5$ ✓

$$\therefore \text{单个 transistor 的最大功耗为 } \frac{V_{CC}^2}{\pi^2 R_L} = 0.91\text{ W}$$

12.8 Consider a V_{BE} multiplier with $R_1 = R_2 = 1.2 \text{ k}\Omega$, utilizing a transistor that has $V_{BE} = 0.6 \text{ V}$ at $I_C = 1 \text{ mA}$, and a very high β . (a) Find the value of the current I that should be supplied to the multiplier to obtain a terminal voltage of 1.2 V . (b) Find the value of I that will result in the terminal voltage changing (from the 1.2-V value) by $+50 \text{ mV}$, $+100 \text{ mV}$, $+200 \text{ mV}$, -50 mV , -100 mV , -200 mV .

Ans. (a) 1.5 mA ; (b) 3.24 mA , 7.93 mA , 55.18 mA , 0.85 mA , 0.59 mA , 0.43 mA



$$(a) I_R = \frac{1.2 \text{ V}}{2.4 \text{ k}\Omega} = 0.5 \text{ mA}$$

$$\text{此时 } V_{BE} = 0.6 \text{ V} \Rightarrow I_C = 1 \text{ mA}$$

$$\therefore I_{BIAS} = 1.5 \text{ mA}$$

$$(b) V_{BB2} = 1.25 \text{ V} \Rightarrow I_{R2} = \frac{1.25 \text{ V}}{2.4 \text{ k}\Omega} = 0.52 \text{ mA}$$

$$\begin{cases} V_{BE1} = V_T \ln \frac{I_{C1}}{I_S} \\ V_{BE2} = V_T \ln \frac{I_{C2}}{I_S} \end{cases}$$

$$\Rightarrow V_{BE2} - V_{BE1} = V_T \ln \frac{I_{C2}}{I_{C1}}$$

$$\therefore 25 \text{ mV} = 25 \text{ mV} \cdot \ln \frac{I_{C2}}{1 \text{ mA}}$$

$$\therefore I_{C2} = 2.72 \text{ mA}$$

$$\therefore I_{BIAS2} = I_{R2} + I_{C2} = 3.24 \text{ mA}$$